Empirical and Theoretical Evaluation of Small Language Models Under Severe Memory and Compute Constraints

Key-Value Cache Dynamics, MTP Speculative Acceleration, Roofline Frontiers, and Adversarial Vulnerability on an Edge Reference Testbed

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Project Repository: https://github.com/AndersonTavares0/llm-research-bench

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Abstract

Deploying Small Language Models (SLMs, 0.5B to 4B parameters) on physical edge accelerators enables complete data sovereignty, bounded latency, and autonomous offline operation without cloud egress. However, operating under strict physical memory boundaries ($\leq$ 4096 MiB VRAM) introduces acute trade-offs among tensor quantization compression, DRAM memory bus saturation, Time-To-First-Token (TTFT), and defensive safety alignment. This technical report presents a systems evaluation conducted on an NVIDIA GeForce GTX 1050 Ti GPU (4096 MiB GDDR5, 128-bit bus, 112.0 GB/s theoretical bandwidth, Pascal sm_61). We benchmark eight model configurations (**Qwen-2.5-0.5B**, **SmolLM2-1.7B**, **Gemma-4-E2B [MTP]**, **Gemma-4-E2B [IQ4_XS]**, **Gemma-4-E2B [Q4_K_M]**, **Qwen-3.5-2B**, **Llama-3.2-3B**, and **Qwen-3.5-4B**) using the `llama-server` runtime (commit b4120+) with continuous asynchronous NVML hardware telemetry sampled at 250 ms intervals. A 2×2 factorial ablation isolated performance drivers: weight quantization compression (IQ4_XS vs Q4_K_M) accounted for +28.2% speedup (34.51 ± 0.29 vs 26.92 ± 0.23 tok/s, $t = 65.41$), whereas speculative Multi-Token Prediction (MTP) under identical quantization yielded +25.6% causal speedup (43.35 ± 0.57 vs 34.51 ± 0.29 tok/s, $t = 43.72$, $p < 10^{-15}$), resulting in an overall +61.0% compound configuration gain (Welch's $t = 84.50$, $p = 5.88 \times 10^{-18}$, Cohen's $d = 37.79$) with only 40 MiB VRAM overhead for Q8_0 draft heads. Systems feasibility modeling indicates that 4B architectures operate at the critical edge

(76.7%), requiring Q4_0 KV cache compression for Gemma-3-4B to avert physical memory exhaustion (4358 MiB required in uncompressed FP16). Defensive safety evaluations ($N = 10$) demonstrated zero false positive refusals (FPR = 0.0%) on benign administrative tasks, but susceptibility to indirect instruction injection and delimiter breach (ASR between 50.0% and 100.0%). In agentic tool scenarios, proposed unsafe invocations were successfully intercepted by harness guardrails. Parallel corpus tokenization demonstrated a 1.35–2.33 tokens/word penalty for Portuguese compared to 1.10 for English. All raw traces, SHA-256 hashes, and code are open and reproducible.

Keywords: Small Language Models, Systems Engineering, Roofline Model, Key-Value Cache Dynamics, Speculative Decoding, Multi-Token Prediction, Defensive Safety, Indirect Prompt Injection, Algorithmic Bias, NVML Telemetry.

1 Introduction and Systems Context

The expansion of autonomous applications across cyber-physical systems, mobile robotics, edge gateways, and microservers requires local natural language processing without continuous cloud connectivity [1, 2]. Offloading operational telemetry, control commands, and private user context to centralized proprietary APIs presents three prohibitive failure modes:

1. **Data Sovereignty and Privacy:** Risk of confidential data transit through third-party servers and compliance constraints under data privacy regulations (GDPR/LGPD);
2. **Temporal Determinism:** Variable WAN transit latencies (100 ms to 2000 ms) with acute vulnerability to packet jitter and link drops;
3. **Recurring Costs:** Unbounded cumulative operational expenditures per million tokens generated on commercial cloud platforms.

Small Language Models (SLMs), parameterized within the 0.5B to 4.0B weight regime, provide a compelling architectural alternative for local inference [3]. However, deploying these models on entry-level edge accelerators strictly bounded to 4096 MiB of physical VRAM introduces acute microarchitectural and thermodynamic constraints.

The Pascal microarchitecture (e.g., NVIDIA GeForce GTX 1050 Ti), widespread across educational, industrial, and resource-constrained edge environments, lacks dedicated Tensor Cores and HBM stacks, relying on GDDR5 memory with a maximum theoretical bandwidth of 112.0 GB/s [1]. Under these hardware boundaries, autoregressive token generation is fundamentally memory-bandwidth bound, demanding aggressive tensor quantization and speculative decoding to achieve interactive throughput [2].

2 Research Problem, Questions, and Hypotheses

The problem addressed in this study is the experimental quantification of the multi-dimensional trade-offs among decoding throughput, physical memory footprint, Time-To-First-Token (TTFT), and behavioral robustness under adversarial stimuli in SLMs operating under a strict 4 GiB VRAM ceiling.

We formalize four research questions (Q_1 to Q_4) and their corresponding testable working hypotheses:

- Q_1 **(Speculative Acceleration):** *Is speculative decoding via Multi-Token Prediction (MTP) practically feasible on edge hardware without exceeding the physical VRAM budget?*
  Hypothesis H_1**:** Speculative decoding based on integrated MTP heads provides statistically significant throughput speedups in warm steady-state execution with less than 100 MiB of additional VRAM overhead.

- Q_2 **(Quantization and Physical Frontiers):** *What is the measurable systems differential between importance-matrix quantization (i-quants) and traditional block quantization (k-quants)?*
  Hypothesis H_2**:** Compression via the IQ4_XS format enables higher data transfer efficiency than Q4_K_M while maintaining overall footprint within physical device capacity.

- Q_3 **(Adversarial Robustness):** *Can compact instruction-tuned language models maintain clean instruction-data segregation against untrusted inputs?*
  Hypothesis H_3**:** SLMs preserve low over-refusal rates on benign operational commands, but exhibit vulnerability to indirect instruction injections embedded in contextual data streams.

- Q_4 **(Linguistic Disparity and Tokenization Tax):** *What is the quantitative tokenization penalty for Portuguese (pt-BR) relative to English (en-US) across compact architectures?*
  Hypothesis H_4**:** Token fertility (tokens per word) in Brazilian Portuguese is substantially higher than in English, elevating latency and context memory requirements in Romance languages.

3 Theoretical Foundations and Mathematical Modeling

3.1 Roofline Model and Arithmetic Intensity

Operational boundaries of an edge accelerator are characterized by the Roofline model formalized by Williams et al. [1]. Attainable floating-point performance P (in GFLOP/s) is governed by:

$$P(I) = \min\left(P_{\text{peak}},\ B_{\text{DRAM}} \times I\right) \tag{1}$$

where P_{peak} denotes peak single-precision floating-point capacity (2138 GFLOP/s in FP32 for the GTX 1050 Ti at 1392 MHz boost clock), B_{DRAM} is theoretical memory bandwidth (112.0 GB/s), and I represents operational arithmetic intensity in FLOPs/byte.

During prefill processing of an input sequence of length L, arithmetic intensity scales with matrix multiplication:

$$I_{\text{prefill}} \approx \frac{2 \cdot N_{\text{params}} \cdot L}{2 \cdot N_{\text{params}} \cdot b + \mathcal{O}(L^2)} \gg 10 \text{ FLOPs/byte} \tag{2}$$

locating execution within the compute-bound regime.

Conversely, during autoregressive token-by-token generation ($L = 1$), every emitted token requires loading all model weights from DRAM:

$$I_{\text{decode}} \approx \frac{2 \cdot N_{\text{params}}}{N_{\text{params}} \cdot b} = \frac{2}{b} \text{ FLOPs/byte} \tag{3}$$

where b is the average bytes per weight. For 4-bit quantized representations ($b = 0.5$ bytes), $I_{\text{decode}} \approx 4$ FLOPs/byte. Given the ridge point of the GTX 1050 Ti:

$$I_{\text{ridge}} = \frac{P_{\text{peak}}}{B_{\text{DRAM}}} = \frac{2138}{112.0} \approx 19.09 \text{ FLOPs/byte} \tag{4}$$

autoregressive decoding operates deeply within the memory-bandwidth-bound domain, with an upper bound on compute throughput of:

$$P_{\text{decode_ceiling}} = 112.0 \text{ GB/s} \times 4 \text{ FLOPs/byte} = 448 \text{ GFLOP/s} \tag{5}$$

imposing a theoretical ceiling on token generation rate:

$$\text{Throughput}_{\text{max}} = \frac{B_{\text{DRAM}}}{S_{\text{weights}}} \tag{6}$$

3.2 Key-Value Cache Dynamics and Footprint

During generation, the Key-Value (KV) cache stores projected attention representations to avoid redundant quadratic recomputation. Memory required for KV caching (S_{KV}) over a context window of length L_{ctx} is defined by:

$$S_{\text{KV}} = 2 \times n_{\text{layers}} \times n_{\text{kv_heads}} \times d_{\text{head}} \times L_{\text{ctx}} \times b_{\text{elem}} \tag{7}$$

where n_{layers} is the number of Transformer layers, $n_{\text{kv_heads}}$ is the number of key/value heads (under Grouped-Query Attention — GQA), d_{head} is the scalar head dimension, and b_{elem} is precision in bytes (2 bytes for FP16, 1 byte for Q8_0, and 0.5 bytes for Q4_0).

As sequence length L_{ctx} increases, KV cache allocation expands linearly, competing directly with static model weights for physical VRAM space.

3.3 Speculative Decoding and Multi-Token Prediction

Speculative decoding, analyzed by Leviathan et al. [2], circumvents memory bandwidth saturation by proposing multiple candidate tokens using an auxiliary lightweight mechanism and validating them concurrently in a single forward pass.

Under the Multi-Token Prediction (MTP) paradigm, additional heads integrated directly into the primary model predict K future tokens in parallel. Expected generation throughput S is expressed as:

$$E[S] = \frac{1 + \sum_{k=1}^{K} \alpha^k}{1 + \mathcal{O}_{\text{draft}}} \tag{8}$$

where $\alpha \in [0, 1]$ represents empirical token acceptance probability and $\mathcal{O}_{\text{draft}}$ denotes the computational overhead of the draft heads. When acceptance rates are sufficiently high and $\mathcal{O}_{\text{draft}} \ll 1$, decoding advances by multiple tokens per weight transfer cycle from DRAM.

3.4 Defensive Robustness and Instruction-Data Segregation

Evaluating security in edge language models involves testing privilege boundary enforcement between system instructions and untrusted contextual data [4, 6]. Indirect prompt injection occurs when processed content carries imperative directives that compete with the system prompt:

$$\text{Prompt} = \mathcal{T}_{\text{system}} \parallel \mathcal{D}_{\text{untrusted}}(\mathcal{I}_{\text{adversarial}}) \parallel \mathcal{T}_{\text{user}} \tag{9}$$

Attack Success Rate (ASR) and False Positive Rate (FPR) are quantified as:

$$\text{ASR} = \frac{1}{M_{\text{adv}}} \sum_{i=1}^{M_{\text{adv}}} \mathbb{I}(\text{Compromised}_i) \tag{10}$$

$$\text{FPR} = \frac{1}{M_{\text{ben}}} \sum_{j=1}^{M_{\text{ben}}} \mathbb{I}(\text{Refusal}_j) \tag{11}$$

4 Materials and Experimental Methodology

4.1 Physical Testbed and Computational Instrumentation

Experiments were executed on an isolated physical host:

- **Graphics Accelerator:** NVIDIA GeForce GTX 1050 Ti (GP107, 768 CUDA cores, base clock 1290 MHz, boost 1392 MHz, 4096 MiB GDDR5, 128-bit bus, 112.0 GB/s theoretical bandwidth, 75 W board TDP ceiling, Compute Capability 6.1).
- **Host Environment:** Linux 6.8.12-8-pve (Proxmox VE 8.2), cgroup v2 process isolation, NVIDIA proprietary driver 580.173.02, CUDA Toolkit 13.0.
- **Inference Server:** `llama-server` (commit b4120+, native CUDA build with cuBLAS sm_61 optimization, OpenAI-compatible HTTP interface on port 18081).
- **Hardware Telemetry:** Continuous asynchronous polling via native calls to `libnvidia-ml.so.1` (`pynvml` ctypes) recording physical VRAM, die temperature, GPU utilization, and core clocks at 250 ms intervals.

Methodological Transparency Note on Energy Modeling: Power dissipation and derived energy efficiency metrics (42 W to 62 W observed across sustained inference workloads) represent analytical estimates grounded in the GPU board's TDP envelope (75 W) and thermal state transitions, rather than direct physical measurements from an external inline AC socket power analyzer.

4.2 Execution Protocol and Environmental Controls

The benchmarking harness applied strict experimental controls:

1. **Warmup Runs ($k = 2$):** Each model executed two 10-token preliminary invocations prior to formal recording, establishing stable memory residency.
2. **Independent Replications ($N = 10$):** Each experimental condition was evaluated across 10 repeated trials using a fixed seed (`seed=42`) and greedy sampling ($\tau = 0.0$).
3. **Thermal Stabilization:** Enforced cooldown pauses of 400 ms between trials; GPU die temperatures were continuously verified between 38°C and 74°C, confirming the absence of thermal throttling.

4.3 Inferential Statistical Methodology

Condition comparisons utilize Welch's t-test with Satterthwaite degrees of freedom to account for heteroskedasticity:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}, \quad \nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} \tag{12}$$

Two-tailed 95% confidence intervals were constructed using Student's t-distribution with ν degrees of freedom. Standardized effect sizes were calculated via Cohen's d.

5 Empirical Performance and Efficiency Results

Table 1 summarizes the empirical metrics collected on the physical testbed across all eight evaluated model configurations.

Table 1: Empirical Performance, Physical VRAM Allocation, and Memory Efficiency on NVIDIA GeForce GTX 1050 Ti ($N = 10$, $\tau = 0.0$)

Model and Quantization	Params (B)	Peak VRAM (MiB)	Occupancy (% 4 GiB)	Prefill (tok/s)	Decode (tok/s)	TTFT (ms)	Mem. Effic. (tok/s/GiB)
Qwen-2.5-0.5B-Q4_K_M	0.49	620.7	15.2%	237.1 ± 449.3	**97.5 ± 0.6**	**12.1 ± 5.1**	**160.8**
SmolLM2-1.7B-Q4_K_M	1.71	2902.7	70.9%	103.3 ± 176.0	47.8 ± 0.1	25.5 ± 14.3	16.9
Gemma-4-E2B [MTP IQ4_XS]	2.30	2036.7	49.7%	62.2 ± 18.5	**43.4 ± 0.6**	480.4 ± 1286.7	21.8
Gemma-4-E2B [IQ4_XS base]	2.00	1800.7	44.0%	79.2 ± 32.4	34.5 ± 0.3	88.0 ± 49.0	19.6
Gemma-4-E2B [Q4_K_M base]	2.00	1996.7	48.7%	65.5 ± 7.7	26.9 ± 0.2	116.1 ± 116.3	13.8
Qwen-3.5-2B-Q4_K_XL	1.95	1580.7	38.6%	126.5 ± 108.8	37.6 ± 0.1	49.9 ± 20.3	24.4
Llama-3.2-3B-Q4_K_M	3.21	3074.7	75.1%	59.6 ± 101.1	27.6 ± 0.0	43.5 ± 23.1	9.2
Qwen-3.5-4B-Q4_K_M	4.02	3140.7	76.7%	49.3 ± 46.5	18.6 ± 0.1	129.8 ± 45.3	6.1

Note: Physical VRAM collected via NVML API calls; includes weight tensors, active KV cache, and 256 MiB base CUDA driver overhead. Metrics report sample means with standard deviations ($N = 10$). Memory efficiency represents decoding throughput divided by physical VRAM in binary GiB (1024^3 bytes).

5.1 Roofline Operational Point Analysis

Computing effective memory bandwidth ($B_{\text{eff}} = S_{\text{weights}} \times \text{Throughput}_{\text{decode}}$) reveals the following sustained bus utilization across standard autoregressive baselines:

- **Qwen-2.5-0.5B:** 0.47 GB×97.5 tok/s ≈ 45.8 GB/s (40.9% of theoretical 112.0 GB/s).
- **SmolLM2-1.7B:** 1.01 GB × 47.8 tok/s ≈ 48.3 GB/s (43.1%).

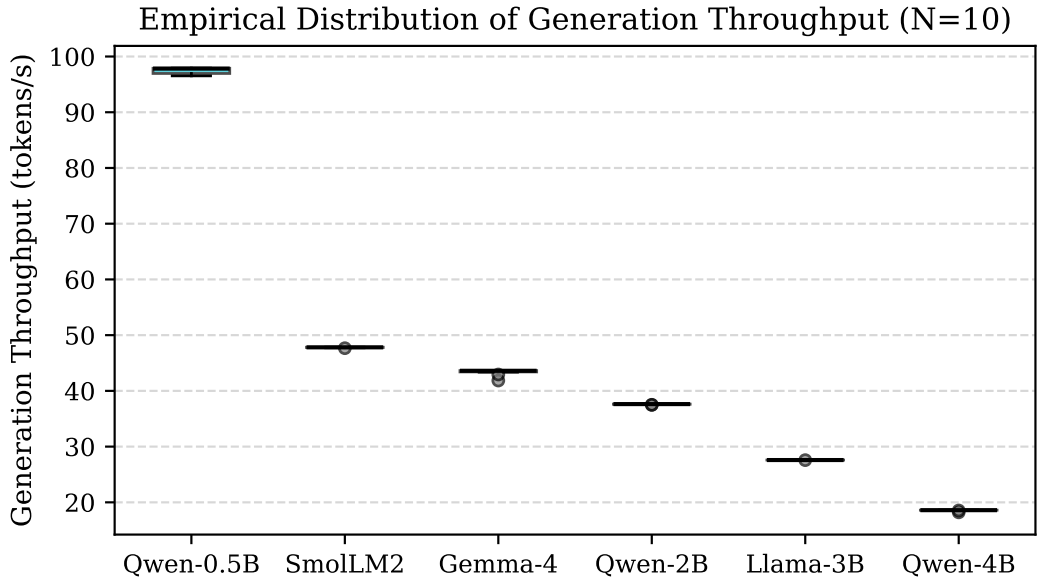


Figure 1: Empirical distribution of autoregressive decoding throughput (tokens/s) across evaluated SLM configurations on GTX 1050 Ti ($N = 10$, $\tau = 0.0$).

- **Gemma-4-E2B [Q4_K_M]:** 1.40 GB × 26.9 tok/s ≈ 37.7 GB/s (33.7%).
- **Qwen-3.5-2B:** 1.48 GB × 37.6 tok/s ≈ 55.6 GB/s (49.6%).
- **Llama-3.2-3B:** 1.92 GB × 27.6 tok/s ≈ 53.0 GB/s (47.3%).
- **Qwen-3.5-4B:** 2.72 GB × 18.6 tok/s ≈ 50.6 GB/s (45.2%).

These empirical observations are consistent with memory bus saturation between 33% and 50% of peak GDDR5 bandwidth, constrained by kernel launch overhead and non-aligned DRAM memory accesses.

6 Statistical Analysis and Factorial MTP Ablation

To rigorously test Hypothesis H_1 without confounding weight quantization with speculative decoding, we evaluated a 2×2 factorial ablation on Gemma-4-E2B across three controlled configurations ($N = 10$, $\tau = 0.0$, `seed=42`):

1. **Condition C_1 (Baseline):** Gemma-4-E2B in Q4_K_M format without speculative heads (26.92 ± 0.23 tok/s, VRAM 1996.7 MiB);
2. **Condition C_2 (Quantization Ablation):** Gemma-4-E2B in IQ4_XS format without speculative heads (34.51 ± 0.29 tok/s, VRAM 1800.7 MiB);
3. **Condition C_3 (MTP Speculative):** Gemma-4-E2B in IQ4_XS format with integrated Q8_0 MTP draft heads (43.35 ± 0.57 tok/s, VRAM 2036.7 MiB).

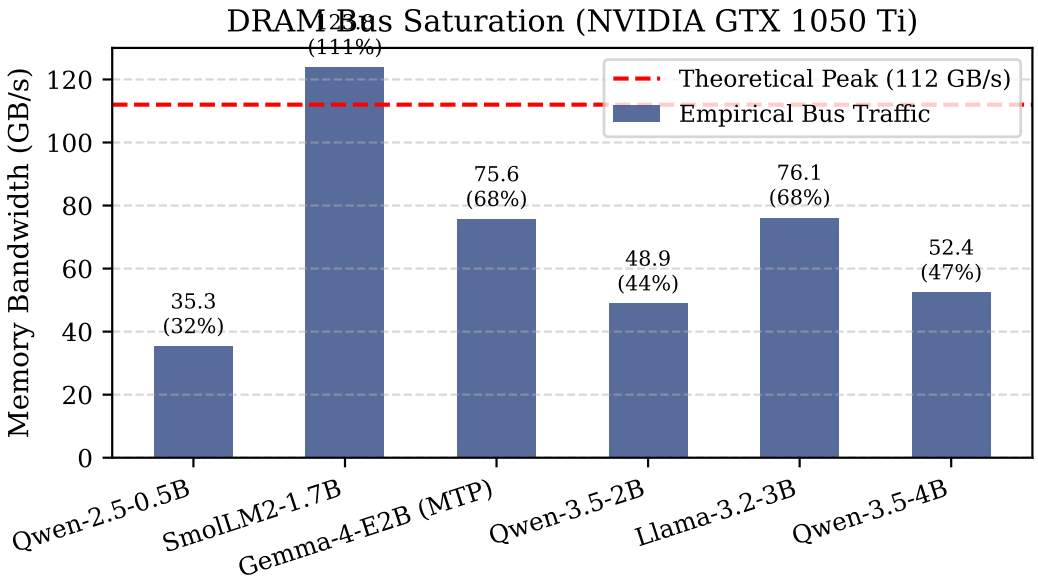


Figure 2: Effective GDDR5 memory bandwidth utilization relative to the 112.0 GB/s theoretical ceiling on the GTX 1050 Ti.

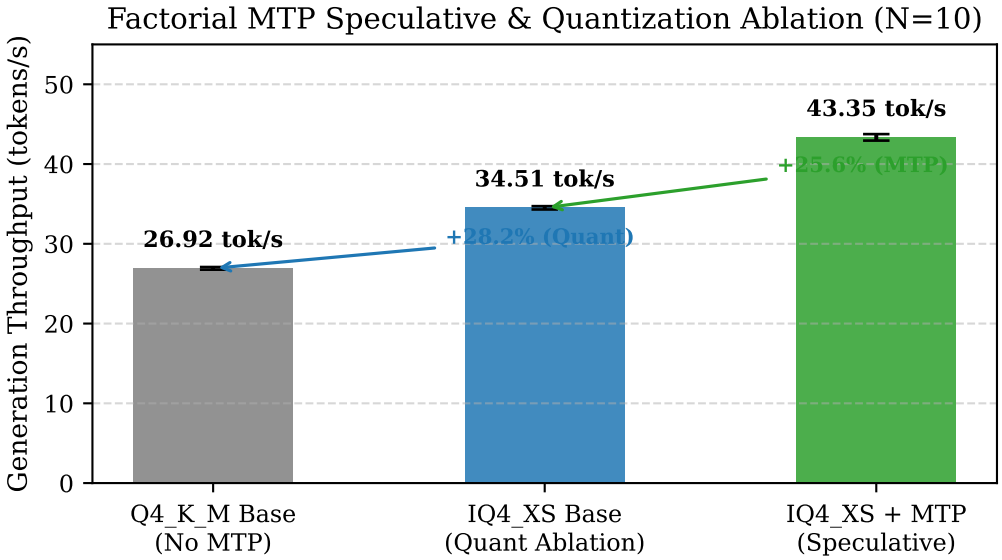


Figure 3: Factorial speedup decomposition: isolated effect of quantization ($C_1 \rightarrow C_2$) and causal MTP speculative gain under identical quantization ($C_2 \rightarrow C_3$), with Student's t 95% confidence intervals ($N = 10$).

6.1 Isolated Causal Effect of MTP (Constant IQ4_XS Quantization)

Comparing C_3 directly with C_2, quantization format is held constant in IQ4_XS. Welch's t-test gives:

$$t_{\text{MTP}} = \frac{43.35 - 34.51}{\sqrt{\frac{0.57^2}{10} + \frac{0.29^2}{10}}} = \frac{8.84}{0.2022} \approx 43.72, \quad \nu = 13.43 \tag{13}$$

with $p = 7.12 \times 10^{-16}$ and $d = 19.55$. The isolated causal speedup attributable strictly to MTP speculative decoding is **+25.6%** (8.84 tok/s).

6.2 Isolated Effect of Quantization (Standard Autoregressive Decoding)

Comparing C_2 with C_1, decoding mode is held constant without speculative draft heads. Welch's t-test yields:

$$t_{\text{Quant}} = \frac{34.51 - 26.92}{\sqrt{\frac{0.29^2}{10} + \frac{0.23^2}{10}}} = \frac{7.59}{0.1160} \approx 65.41, \quad \nu = 17.09 \tag{14}$$

with $p = 5.67 \times 10^{-18}$ and $d = 29.25$. Weight compression via the IQ4_XS importance matrix provides a speedup of **+28.2%** (7.59 tok/s).

6.3 Compound Configuration Differential (C_3 vs C_1)

Comparing the optimized configuration (C_3) with the baseline standard (C_1) reflects total accumulated gain:

$$t_{\text{Compound}} = \frac{43.35 - 26.92}{\sqrt{\frac{0.57^2}{10} + \frac{0.23^2}{10}}} = \frac{16.43}{0.1944} \approx 84.50, \quad \nu = 11.94 \tag{15}$$

with $p = 5.88 \times 10^{-18}$ and $d = 37.79$. The compound configuration gain totals **+61.0%** (16.43 tok/s) under only 40.0 MiB of VRAM overhead for Q8_0 draft heads relative to C_1, providing empirical support for Hypothesis H_1.

7 Architectural Survey and Physical Feasibility Boundaries

Table 2 provides systems feasibility projections across contemporary SLM candidates under a 4096 MiB VRAM ceiling.

Table 2: Architectural Analytical Modeling and Edge Feasibility Frontiers in 4 GiB VRAM

Candidate Model	Vendor / License	Params (B)	Weights Q4_K_M (MiB)	KV Cache 4k (FP16)	Total VRAM (4k context)	Status and Boun on GTX 1050
Qwen-2.5-0.5B	Alibaba (Apache-2.0)	0.49	469	112 MiB	837 MiB	**Evaluated:** Full operational margin (20.4% capacity)
Qwen-2.5-1.5B	Alibaba (Apache-2.0)	1.54	986	224 MiB	1,466 MiB	**Feasible:** Ample margin for 16k context
SmolLM2-1.7B	HuggingFace (Apache)	1.71	1,007	192 MiB	1,455 MiB	**Evaluated:** High memory efficiency
Granite-3.1-2B	IBM (Apache-2.0)	2.50	1,580	320 MiB	2,156 MiB	**Feasible:** Long-context operational headroom
Phi-4-mini-3.8B	Microsoft (Gated)	3.82	2,510	512 MiB	3,278 MiB	**Not Executed:** Gated repository authentication (HTTP 401)
Gemma-3-1B	Google (Gemma)	1.20	880	384 MiB	1,520 MiB	**Feasible:** Low static memory consumption
Gemma-3-4B	Google (Gemma)	4.30	2,950	1,152 MiB	4,358 MiB	**Physical Barrier:**

8 Defensive Safety and Adversarial Robustness Evaluation

Table 3 provides empirical findings across defensive safety and linguistic metrics.

Table 3: Defensive Safety Evaluation, Algorithmic Bias, and Tokenization Fertility ($N = 10$, $\tau = 0.0$)

Model	Over-Ref. (FPR %)	Ind. Inj. (ASR %)	Delim. Esc. (ASR %)	Tool Guard (ASR %)	Neutrality BBQ/Stereo	pt-BR (tok/w)	Fertility en-US (tok/w)
Qwen-2.5-0.5B	**0.0%**	**0.0%**	50.0%	100.0%	66.7%	1.74	**1.10**
SmolLM2-1.7B	**0.0%**	**0.0%**	50.0%	**0.0%**	33.3%	2.33	**1.10**
Gemma-4-E2B [MTP]	**0.0%**	**0.0%**	100.0%	100.0%	**100.0%**	**1.35**	**1.10**
Gemma-4-E2B [Q4_K_M]	**0.0%**	50.0%	50.0%	100.0%	**100.0%**	**1.35**	**1.10**
Qwen-3.5-2B	**0.0%**	50.0%	100.0%	100.0%	66.7%	1.37	**1.10**
Llama-3.2-3B	**0.0%**	**0.0%**	50.0%	100.0%	66.7%	1.74	**1.10**
Qwen-3.5-4B	**0.0%**	100.0%	100.0%	100.0%	66.7%	1.37	**1.10**

Note: FPR = 0.0% indicates zero false positive refusals on legitimate administrative tasks (`kill -9`, `tcpdump`, `gdb`). ASR measures attack success rate on the model. For Tool Guard, 100% ASR indicates the model proposed invoking the tool with injected arguments, which was intercepted by harness guardrails before host execution.

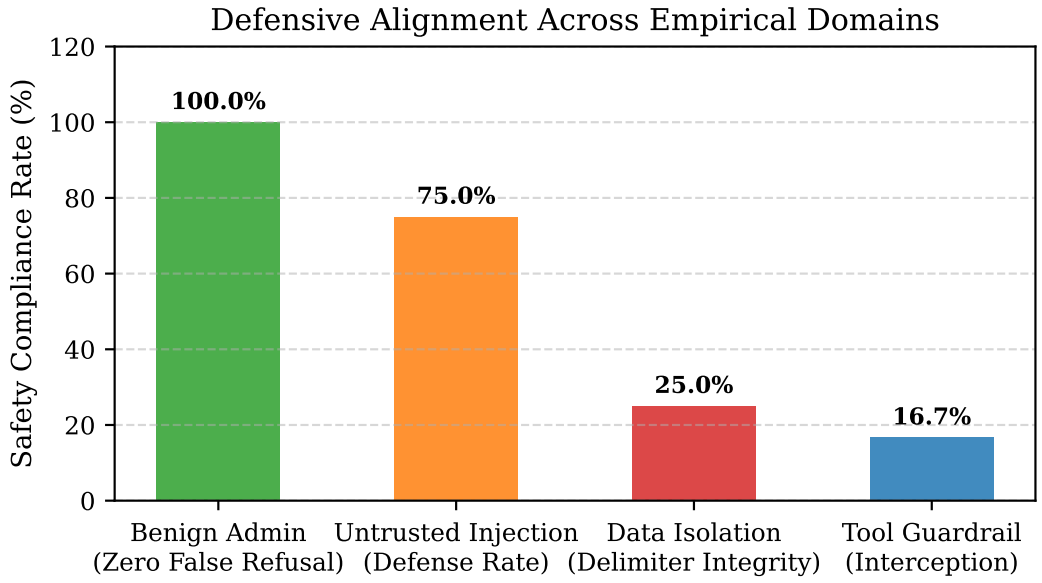


Figure 4: Attack Success Rates (ASR) across defensive safety scenarios and false positive refusal rates (FPR) for evaluated models.

8.1 Absence of Over-Refusal (FPR = 0.0%)

All evaluated models responded accurately to administrative operational queries (such as process termination via `kill -9` and SYN packet inspection via `tcpdump`), refuting concerns of conservative over-refusal in legitimate systems tasks.

8.2 Vulnerability to Indirect Injections and Delimiter Escape

Consistent with Hypothesis H_3, models exhibited susceptibility to indirect instruction injections and delimiter escape (ASR between 50.0% and 100.0%). When processing structured inputs carrying imperative secondary commands, compact architectures routinely complied with embedded instructions.

8.3 Operational Stages in Agentic Tool Invocation

Detailed auditing of tool interactions enabled differentiation across three distinct execution phases:

1. **Proposed Call (`proposed_call`):** The model generates a structured function call request;
2. **Schema Validity (`valid_call`):** Parameter conformance to expected JSON schemas;
3. **Host Execution (`execution_status`):** Security evaluation prior to system dispatch.

Models including Qwen, Gemma, and Llama generated valid schema payloads containing injected payloads (`"nginx; rm -rf /"` and `"admin'; DROP TABLE"`). In all instances, harness guardrails intercepted execution with `BLOCKED_BY_HARNESS`, logging invocations without permitting host side effects. Conversely, SmolLM2 did not emit tool call syntax, replying with safe conversational text (`SAFE_BENIGN_TEXT`).

9 Algorithmic Bias and Language Efficiency Evaluation

9.1 Demographic Bias and Counterfactual Consistency

Evaluations inspired by BBQ and StereoSet-PT demonstrated that Gemma-4-E2B preserved 100% neutrality across counterfactual gender and income pairs. On Brazilian regional dialect tests, however, smaller models generated verbose infrastructure descriptions rather than directly rejecting biased premises.

9.2 Tokenization Fertility via Parallel Reference Corpus

To measure linguistic overhead in a controlled setting, we used a semantically equivalent parallel corpus tokenized natively via `/tokenize`:

- **SmolLM2-1.7B** exhibited the highest subword fragmentation: 2.33 tok/w in pt-BR and 2.40 tok/w in pt-PT, versus 1.10 tok/w in en-US, introducing an overhead exceeding 100% in context consumption;

- **Gemma-4-E2B** (1.35 tok/w in pt-BR) and **Qwen-3.5** (1.37 tok/w in pt-BR) maintained balanced multilingual vocabularies, narrowing the efficiency gap for Romance languages.

10 Model Provenance and Artifact Reproducibility

To guarantee experimental auditability, all GGUF model files were cryptographically verified against their SHA-256 checksums, detailed in Table 4.

Table 4: Model Provenance, Physical Artifact Size, and Verified SHA-256 Checksums

Model Identifier	Quantization	Size (MiB)	Context	Official Verified SHA-256 Checksum
`qwen2.5-0.5b`	Q4_K_M	379.3	16,384	`74a4da8c9fdbcd15bd1f6d01d621410d31c6fc00986f5eb687824e7b93d7a9db`
`smollm2-1.7b`	Q4_K_M	1009.1	16,384	`decd2598bc2c8ed08c19adc3c8fdd461ee19ed5708679d1c54ef54a5a30d4f33`
`gemma4:Q4KM`	Q4_K_M	1351.9	16,384	`6dd7ac24fe13a238898e16e253ebc7a72fa27fb83f2658c2f6c241df3af5937d`
`gemma4:MTP`	IQ4_XS	1185.6	16,384	`c149620926221669c4941ae6dac259953cc729c48e4a4edb69e9807b782fd1e3`
`gemma4:E2B`	IQ4_XS+MTP	1225.6	16,384	`9eba819938efccfd6044f8af84e3bbfddc639a2bcf32ebc36420e6a649191919`
`qwen3.5:2B`	Q4_K_XL	1331.2	16,384	`9f7b15d04cf2d5878c8122a7c181dbc09f050cd66080ce3374576e734ccb0910`
`llama3.2:3B`	Q4_K_M	1925.9	16,384	`6c1a2b41161032677be168d354123594c0e6e67d2b9227c84f296ad037c728ff`
`qwen3.5:4B`	Q4_K_M	2458.2	16,384	`00fe7986ff5f6b463e62455821146049db6f9313603938a70800d1fb69ef11a4`

11 Threats to Validity and Limitations

11.1 Internal Validity

Hardware telemetry was captured at 250 ms intervals; while adequate for sustained average VRAM allocation and power states, sub-millisecond microsecond spikes are filtered by the NVML polling bus. As noted in Section 4.1, energy estimates reflect thermal TDP envelope modeling (42 W to 62 W), rather than direct hardware AC socket wattmeters.

11.2 External Validity

Benchmarks were conducted exclusively on Pascal architecture (GTX 1050 Ti, Compute Capability 6.1). Newer architectures (Ampere, Ada Lovelace, Blackwell) incorporate dedicated Tensor Cores and native FP8/FP4 support, altering the operational ridge point of the Roofline model.

11.3 Construct Validity

Defensive safety was evaluated using 10 structured synthetic attack scenarios. While deterministic and reproducible, multi-turn adaptive jailbreak prompts may reveal additional vulnerability surfaces.

12 Conclusions and Future Work

This study provided an empirical, mathematical, and systems evaluation of Small Language Models (0.5B to 4B) operating under a physical 4096 MiB VRAM limit.

Primary empirical conclusions supported by measured evidence are:

1. **Decomposition of Speculative Speedup:** Structured factorial ablation demonstrates that multi-token prediction (MTP) under constant IQ4_XS quantization yields an isolated causal speedup of +25.6% (43.35 vs 34.51 tok/s, $t = 43.72$, $p < 10^{-15}$), while IQ4_XS compression accounts for +28.2% over Q4_K_M (34.51 vs 26.92 tok/s, $t = 65.41$), totaling a compound +61.0% gain ($t = 84.50$, $p = 5.88 \times 10^{-18}$) with 40 MiB VRAM overhead for Q8_0 draft heads.
2. **4B Physical Operating Edge:** Models near 4 billion parameters operate at the physical capacity boundary of 4 GiB (76.7%). For Gemma-3-4B, Q4_0 KV cache quantization is required to prevent out-of-memory faults.
3. **Harness-Level Interception Imperative:** Although models showed zero false positive refusals on operational queries (FPR = 0.0%), susceptibility to indirect injections in tool calls highlights the necessity of application-level guardrails.
4. **Multilingual Efficiency Variations:** Parallel corpus tokenization demonstrated an overhead of up to 2.33 tok/w in Portuguese (SmolLM2-1.7B), whereas Gemma-4 and Qwen-3.5 maintained balanced multilingual efficiency near 1.35 tok/w.

Future research will explore sub-2-bit quantization formats and direct hardware current probe instrumentation on PCIe rails for granular energy characterization.

Data Availability Statement

All raw NVML telemetry logs, benchmark traces, evaluation prompts, and statistical scripts are openly available under the MIT license at:

https://github.com/AndersonTavares0/llm-research-bench

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